HANDS-ON LAB 02 GUIDE & COPY-PASTE PROMPTS • OMNICHANNEL RETAIL EDITION (RETAIL PULSE)

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TRYOLABS • GOOGLE ANTIGRAVITY WORKSHOP • LAB 02 (EN)

Lab 02: Workspace Rules (`GEMINI.md`) & Progressive Disclosure Skills (`SKILL.md`)

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Language / Idioma: English | Español • **Prompt Playbook:** PROMPTS.md | PROMPTS.es.md • **PDF:** EN (Dark) | EN (Light) | ES (Dark) | ES (Light)

1. Scenario & Architectural Goal

In Lab 01, you manually instructed Antigravity to fix binary **`float`** pricing errors, GS1 GTIN-14 validation, and POS webhook lag ATP drift.

In an enterprise retail engineering team, **you should never have to repeat domain standards in every prompt**.

In **Lab 02**, you will transform Antigravity into a permanent **Retail & Omnichannel Commerce Domain Expert** using two core Antigravity Customization primitives:

1. **Hierarchical Workspace Rules (`GEMINI.md` & `.agents/rules/retail-standards.md`)**:

- Automatically loaded into context to enforce **`Decimal(ROUND_HALF_UP)`** currency math, GS1 GTIN-14 Modulo-10 check-digit validation (**`gtin_14`**), **`contract_min_margin_floor_pct`** protection, and the True ATP formula (**`on_hand - reserved - safety_stock - unprocessed_webhook_delta_qty`**).

2. **Progressive Disclosure Skills (`.agents/skills/margin-otif-audit/SKILL.md`)**:

- Lightweight YAML frontmatter (**`name`**, **`description`**) is indexed by Antigravity without bloating the token window.
- When a developer or commercial buyer asks for a **Margin & Supplier OTIF Audit**, Antigravity dynamically loads **`SKILL.md`**, executes **`scripts/run_otif_margin_audit.py`**, cross-references SQLite **`vendor_contracts`** with **`EDI_856_ASN_AUDIT`** events in **`data/logs/pos_wms_webhooks.jsonl`**, and produces **`data/exports/executive_margin_otif_audit.json`** and **`.md`** (**`$98,250.00`** in recoverable chargebacks across 6 breaching suppliers!).

2. Lab Setup

Run the Lab 02 installer (which automatically applies the Lab 01 solution first so latecomers are caught up!):

```
./labs/lab-02-rules-and-skills/install.sh
```

3. Step-by-Step Resolution Using Google Antigravity

Open PROMPTS.md and execute the prompts in order:

Step 3.1: Generate `GEMINI.md` & `.agents/rules/retail-standards.md`

Use the **Workspace Rules Prompt** in **`PROMPTS.md`** (or **`/plan`**) to create:

- **`GEMINI.md`** at the repository root (**`>= 300`** characters) documenting the database tables (**`sku_catalog`**, **`wms_inventory`**, **`vendor_contracts`**, **`promo_campaigns`**, **`omnichannel_orders`**) and non-negotiable guardrails (**`Decimal`**, **`ROUND_HALF_UP`**, **`gtin_14`**, **`contract_min_margin_floor_pct`**, **`unprocessed_webhook_delta_qty`**).
- **`.agents/rules/retail-standards.md`** documenting financial precision, GS1/RFID standards, True ATP, and Vendor OTIF (**`otif_target_pct`**) chargeback rules.

Step 3.2: Create the Progressive Disclosure Skill (`.agents/skills/margin-otif-audit/SKILL.md`)

Create **`.agents/skills/margin-otif-audit/SKILL.md`** with required YAML frontmatter:

```
---
name: margin-otif-audit
description: >-
  Use this skill when asked to audit commercial gross margin leakage, verify True
  Available-to-Promise (ATP) VIP oversell exposure, or calculate recoverable vendor
  OTIF contract chargeback penalties backed by EDI 856 ASN logs.
---
```

And step-by-step instructions to run **`./.venv/bin/python scripts/run_otif_margin_audit.py`** and inspect **`data/exports/executive_margin_otif_audit.json`** and **`.md`**.

Step 3.3: Trigger the Skill & Verify

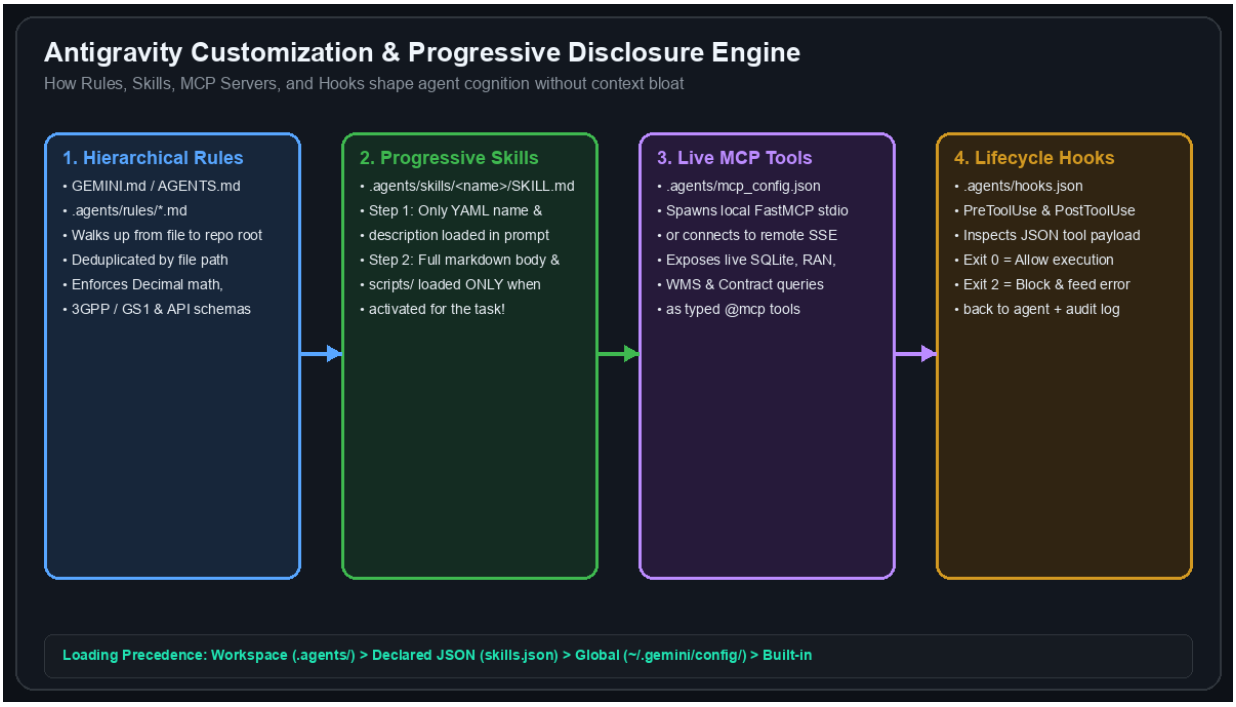
Ask Antigravity in chat:

"Run a full executive margin, True ATP, and supplier OTIF chargeback audit using our workspace skill."

Then run the verification script:

```
./labs/lab-02-rules-and-skills/verify.sh
```

Reference Solution: Run **`./labs/lab-02-rules-and-skills/apply_solution.sh`** at any time.


Antigravity Customization & Progressive Disclosure Engine
How Rules, Skills, MCP Servers, and Hooks shape agent cognition without context bloat
1. Hierarchical Rules
• GEMINI.md / AGENTS.md
• .agents/rules/*.md
• Walks up from file to repo root
• Deduplicated by file path
• Enforces Decimal math,
• 3GPP / GS1 & API schemas
2. Progressive Skills
• .agents/skills/<name>/SKILL.md
• Step 1: Only YAML name &
• description loaded in prompt
• Step 2: Full markdown body &
• scripts/ loaded ONLY when
• activated for the task!
3. Live MCP Tools
• .agents/mcp_config.json
• Spawns local FastMCP stdio
• or connects to remote SSE
• Exposes live SQLite, RAN,
• WMS & Contract queries
• as typed @mcp tools
4. Lifecycle Hooks
• .agents/hooks.json
• PreToolUse & PostToolUse
• Inspects JSON tool payload
• Exit 0 = Allow execution
• Exit 2 = Block & feed error
• back to agent + audit log
Loading Precedence: Workspace (.agents/) > Declared JSON (skills.json) > Global (~/.gemini/config/) > Built-in

LAB 02 COPY-PASTE ANTIGRAVITY PROMPTS (EN)
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Lab 02 — Optimized Antigravity Prompts Playbook (English)

Language / Idioma: English | Español • **Lab Guide:** README.md | README.es.md • **PDF:** EN | ES

Use these copy-pasteable prompts inside **Google Antigravity** to complete Lab 02.

1. Discovery Prompt (Explore Domain & Audit Script)

```
Inspect @src/models.py, @src/services/vendor_otif_auditor.py, @scripts/run_otif_margin_a \
  udit.py, and @tests/test_lab02_rules_skills.py.
Summarize the exact requirements that GEMINI.md, .agents/rules/retail-standards.md, and  \
  .agents/skills/margin-otif-audit/SKILL.md must satisfy.
```

2. `/plan` Prompt (Design Workspace Rules & Progressive Disclosure Skill)

```
/plan Plan the creation of our Antigravity workspace customizations for Retail Pulse:
1. Root GEMINI.md (>= 300 chars) documenting our SQLite schema and non-negotiable engine \
  ering/commercial rules: Decimal with ROUND_HALF_UP, GS1 gtin_14 Modulo-10 validation,  \
  contract_min_margin_floor_pct clamping, True ATP formula deducting unprocessed_webhook \
  _delta_qty, and VIP DIAMOND/GOLD prioritization.
2. .agents/rules/retail-standards.md reinforcing Decimal(ROUND_HALF_UP), gtin_14/RFID ru \
  les, True ATP, and Vendor OTIF (otif_target_pct) chargeback formulas.
3. .agents/skills/margin-otif-audit/SKILL.md with valid YAML frontmatter (name: margin-o \
  tif-audit and third-person description) and step-by-step instructions to execute scrip \
  ts/run_otif_margin_audit.py and inspect data/exports/executive_margin_otif_audit.json  \
  and .md.
```

3. Implementation Prompt (Create Rules & Skill Files)

```
Create GEMINI.md, .agents/rules/retail-standards.md, and .agents/skills/margin-otif-audi \
  t/SKILL.md according to the plan so they satisfy all assertions in @tests/test_lab02_r \
  ules_skills.py.
```

4. Skill Activation & Verification Prompt

```
Activate our margin-otif-audit skill to generate the executive margin, True ATP, and ven \
  dor OTIF audit report. Summarize the 6 breaching vendors and total recoverable chargeb \
  ack USD, then run ./labs/lab-02-rules-and-skills/verify.sh.
```

5. Power-User & Stretch Prompts (`/learn`, `/schedule`, `/compact`)

5A. Auto-Generate Rules with `/learn`

```
/learn Whenever writing a new endpoint or service in Retail Pulse, always return monetar \
  y fields quantized via Decimal("0.01") and cross-check any vendor contract status agai \
  nst EDI_856_ASN_AUDIT records in data/logs/pos_wms_webhooks.jsonl.
```

5B. Recurring Executive OTIF Audit (`/schedule`)

```
/schedule Run scripts/run_otif_margin_audit.py every 30 minutes and alert me if total_re \
  coverable_chargeback_usd exceeds $90,000 or if any DIAMOND order falls into AT_RISK_ST \
  OCKOUT.
```